



PYTHON WEATHER DATA ANALYSER

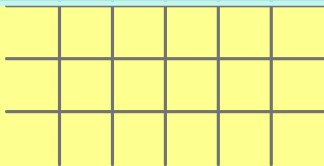


A Hot Summer Morning in Jaipur



Priya hears her mother call out, "It's 42 degrees outside today!"
Priya wonders — how can she check the weather herself?

 Your Mission: Build a Python program to check and analyse weather data!





What is Python?

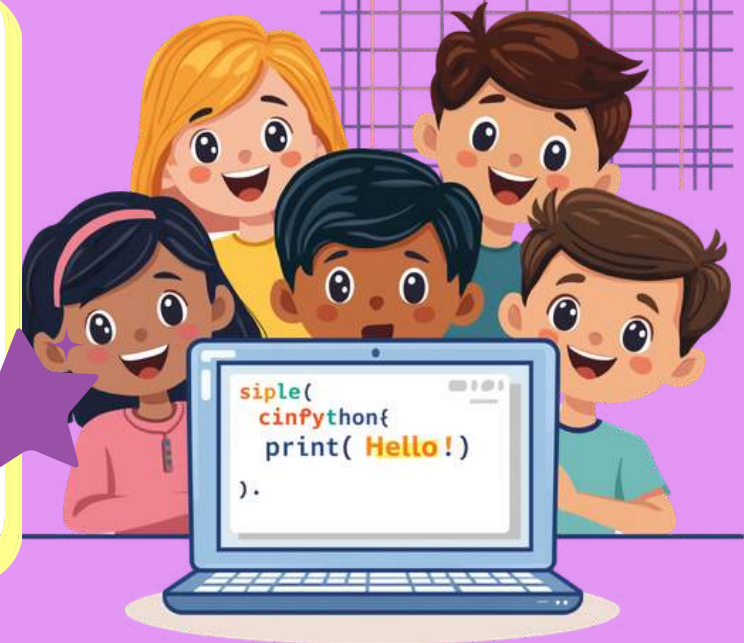
A LANGUAGE FOR COMPUTERS!

Python is a special language we use to give instructions to computers! Just like Amma follows her sambar recipe step-by-step, Python follows our code step-by-step. Big Indian companies like ISRO, Zomato, Swiggy, and IRCTC use Python! Fun fact: It's named after a comedy show called Monty Python, not the snake!



Why Python is Great for Beginners

- Reads almost like English — easy to understand!
- Simple to write — no complicated symbols
- Used by big companies like Google, Netflix, and Instagram
- Perfect first language for young coders like you!



What are Variables?



Variables are like labelled boxes that hold information!
Think of Amma's masala dabba — each container has a label: Turmeric, Cumin seeds, Chilli / Pepper 🌶 Variables work the same way!

Examples:

city = "Mumbai"

temperature = 38





Writing Variables in Python

Use underscores for spaces: city_name
= means "store this value"

Example code:

```
city = "Delhi"
```

```
temperature = 45
```

Remember: = is not math equals, it stores values!





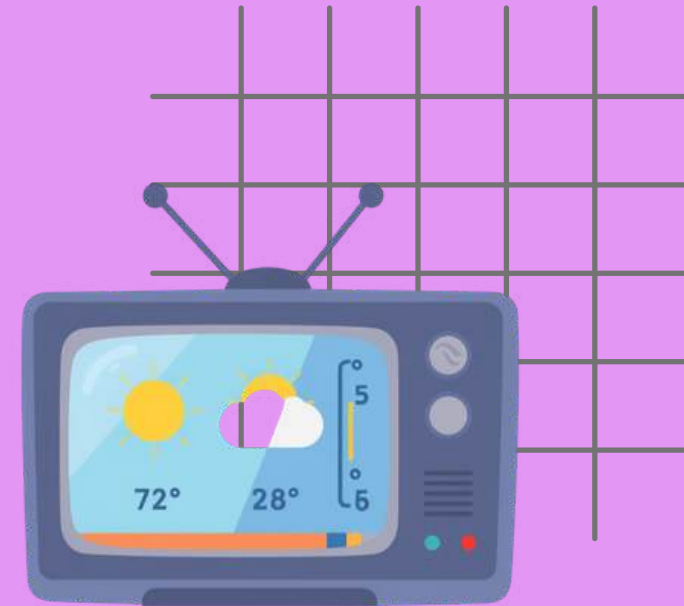
The print() Function

print() displays text or numbers on screen — like a TV weather report!

Examples:

- print("Today's weather:")
- print(temperature)
- print("Temperature:", temperature, "degrees Celsius")

Remember: Use quotes for text!



Indian City Weather Examples



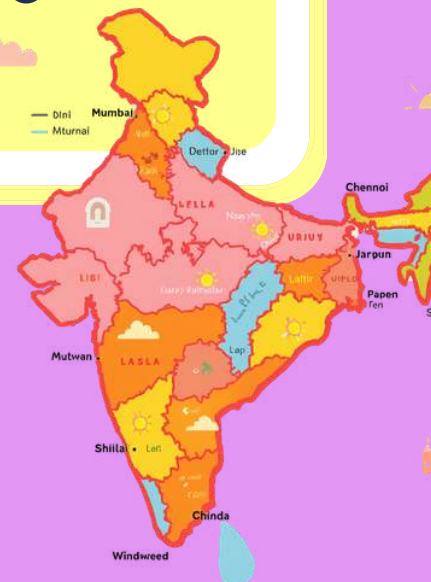
Mumbai: 38°C

Delhi summer: 45°C

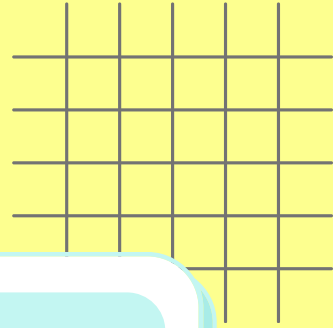
Shimla summer: 18°C

Chennai monsoon: 28°C

Jaipur: 42°C



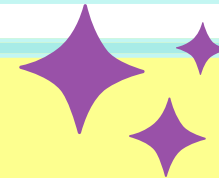
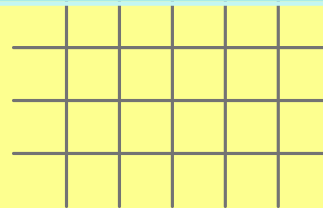
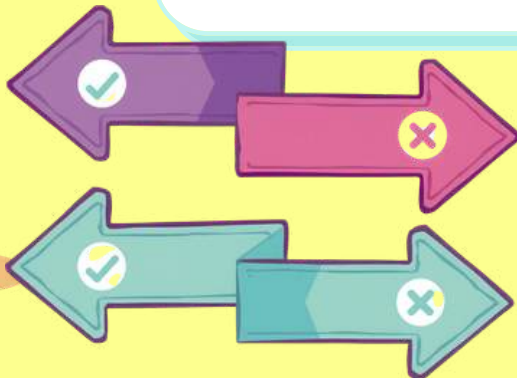
Understanding If-Else Decisions



Computers can make decisions too! If-Else helps your program choose what to do.

- IF something is true → do this action
- ELSE (if not true) → do something different

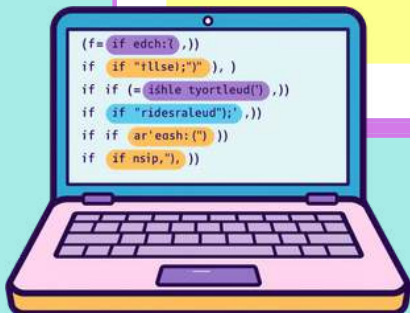
It's like your parents deciding: IF it's very hot outside, wear a cap and carry water. ELSE, you can play normally!



If-Else in Python Code

```
if temperature > 35:  
    print("Hot day! Drink plenty of water.")  
else:  
    print("Cool day! Enjoy the weather.")
```

Remember: Colon : is mandatory after if and else. Indent lines to show blocks. Use == to compare, = to store values.



How If-Else Helps Our Program

WHY DECISIONS MATTER

If-else makes your program smart! Instead of just showing numbers, your Weather Analyser can give helpful advice like "Drink water!" or "Enjoy the cool breeze!" based on the temperature. This makes coding interactive and fun for everyone who uses your program!

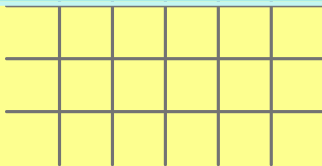


The IPO Model

Think of a juice stall! You give fruit (Input), the juicer blends it (Process), and you get yummy juice (Output)!

Programs work the same way:

- Input → Data goes in
- Process → Computer works on it
- Output → Results come out





Build It! Starter Code

Try filling in the blanks below! Code along with your teacher:

```
city = "_____"  
temperature = _____  
print("Weather in", city)  
print("Temperature:", temperature, "degrees")  
if temperature > 35:  
    print("")  
else:  
    print("")
```



Extension Challenge



Ready for a challenge? Add a third city to your weather program!

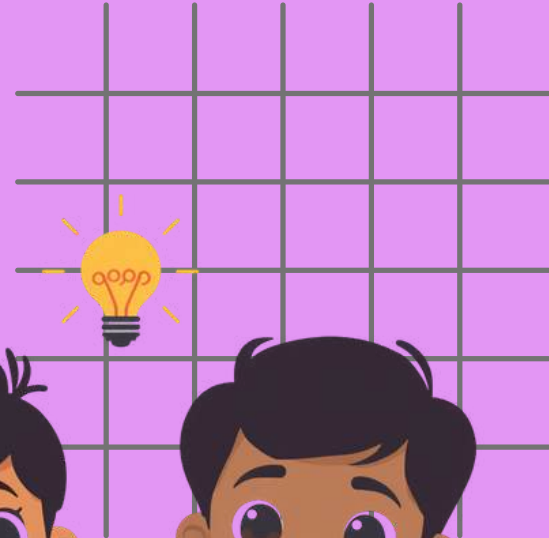
Try this:

- Pick any Indian city you like
- Create a new variable with its temperature
- Use `print()` to show the weather

Hint: `city3 = "Bengaluru"`

`temp3 = 25`

Be creative and explore!



Success Celebration!

Great job! Your Python Weather Analyser works! You've built your very first weather program that can check temperatures and give smart advice. You're now a real coder! Keep exploring, keep creating, and remember — every big programmer started just like you!





What You Learned Today



What Python is and where it is used
Variables and print() function
If-else statements for decisions
Writing simple programs
Running your own code!





Keep Coding and Exploring!

YOUR CODING JOURNEY CONTINUES!

Practice more weather data projects or try fun new ideas! Share your code with friends and family. Remember, coding is creative and exciting - every program you write makes you a better coder. Keep experimenting!





THANK YOU!

ANY QUESTIONS?

PLANRS ACADEMY

